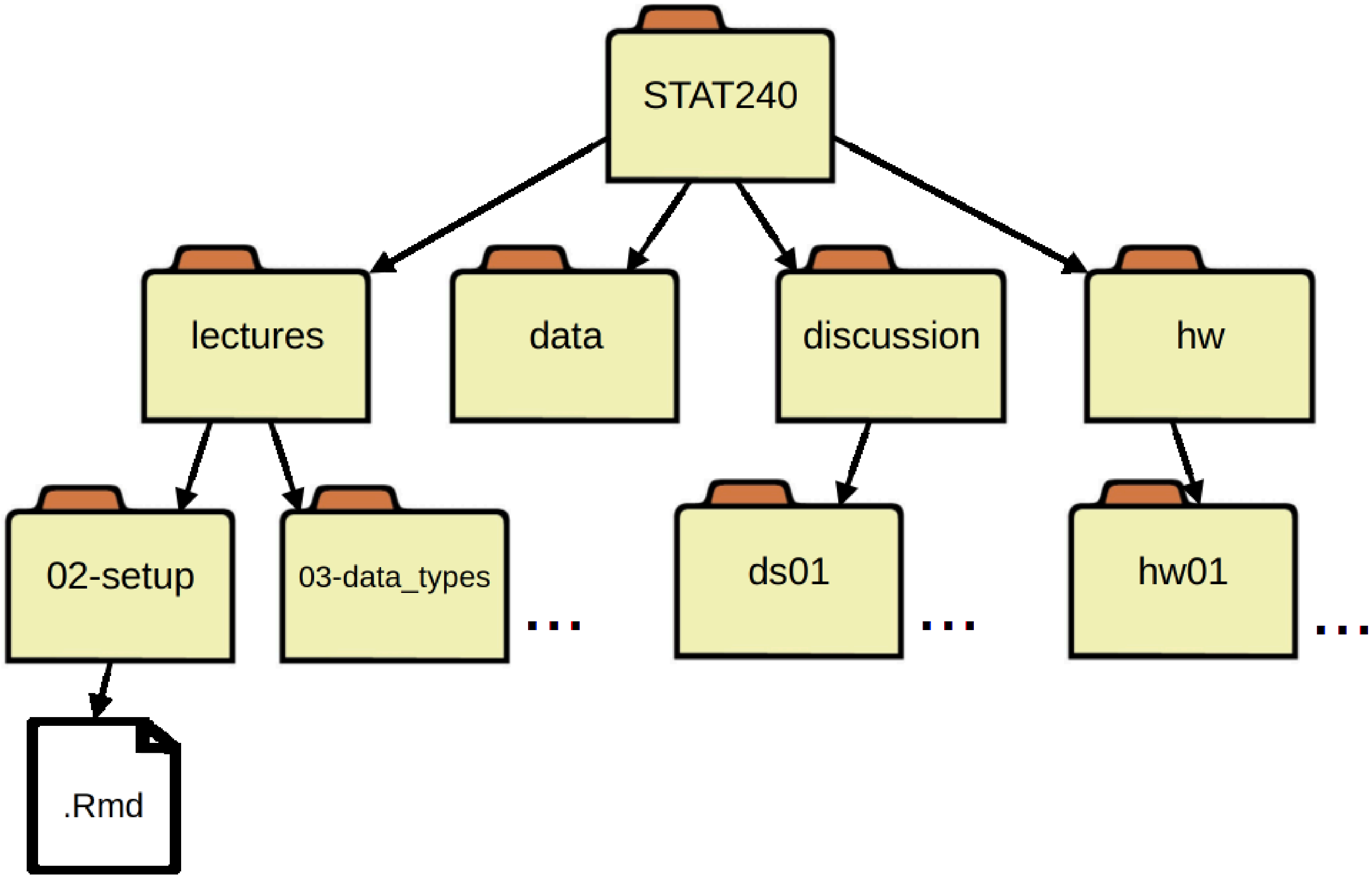
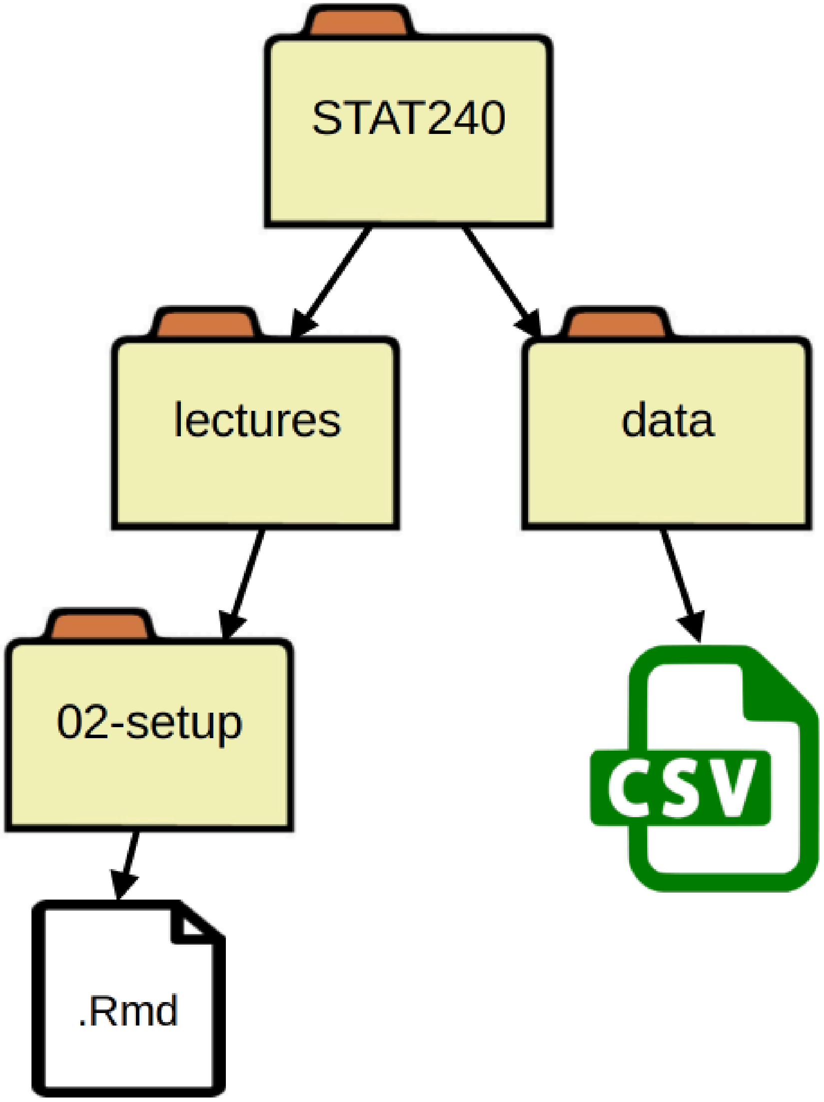
Class Setup

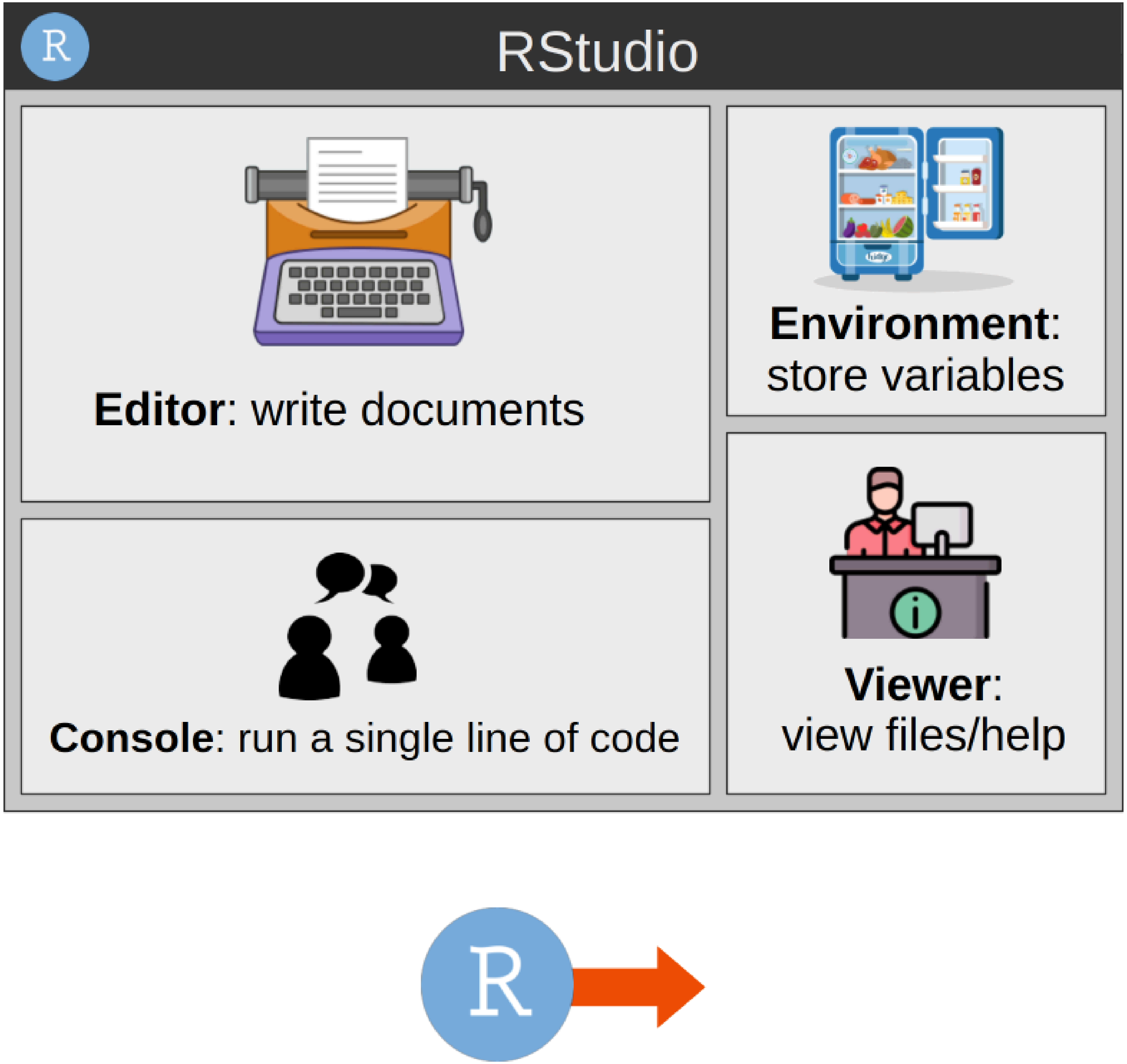
Organization + Intro to R

All of the files (handouts, homework, data, etc.) for 240 will live in folders organized like so:




```r
df <- read_csv("../../data/example.csv")
```


STAT240
lectures
data
02-setup
.Rmd
CSV

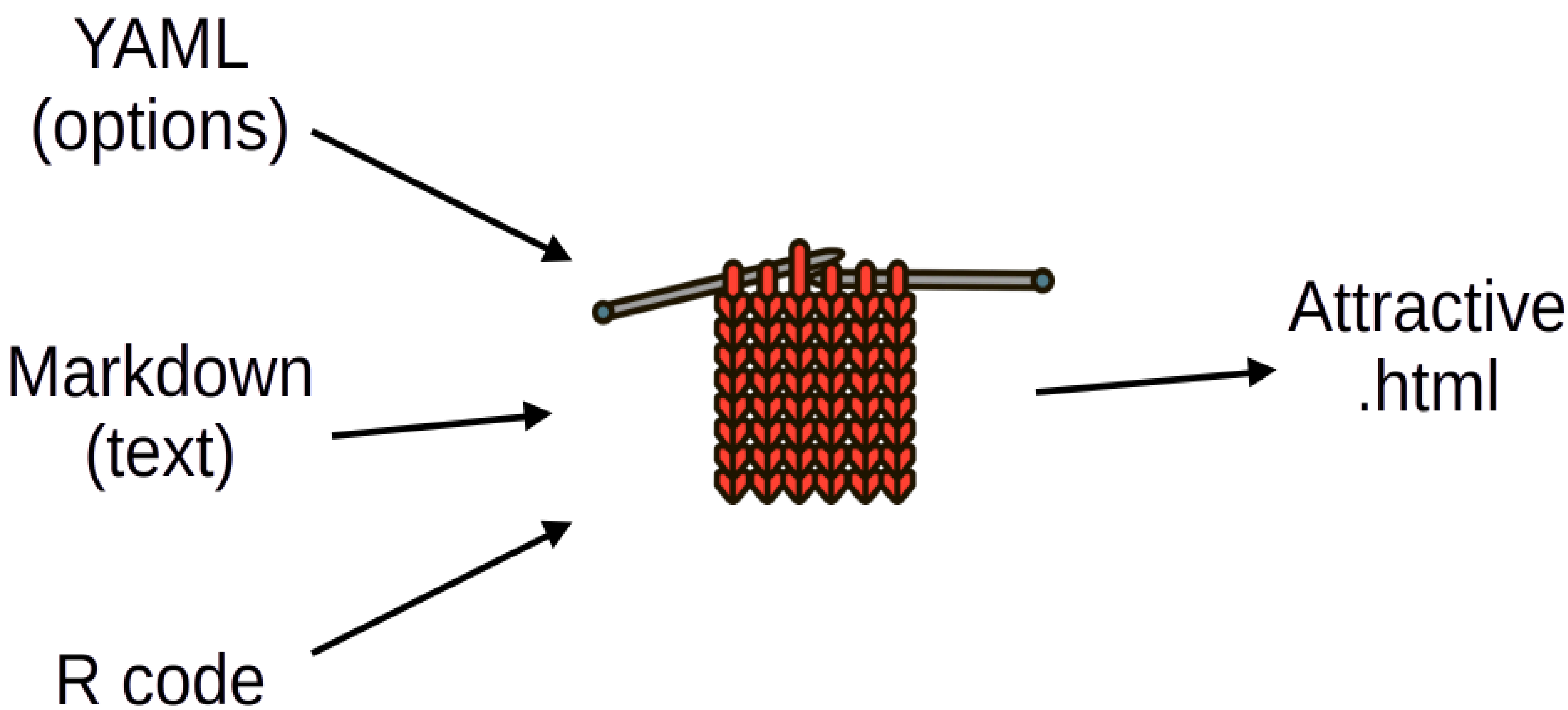

RStudio
Editor: write documents
Environment: store variables
Console: run a single line of code
Viewer: view files/help

R/RStudio questions always welcome at office hours or GLC!

Exercises | Solutions/Hints

- Perform a mathematical operation with variables (not just numbers)
- Clear your console output - can you still get the old code back?
- Find the Apperance Options menu and change the color scheme
- Figure out how to find your R version (it's a pre-defined object)

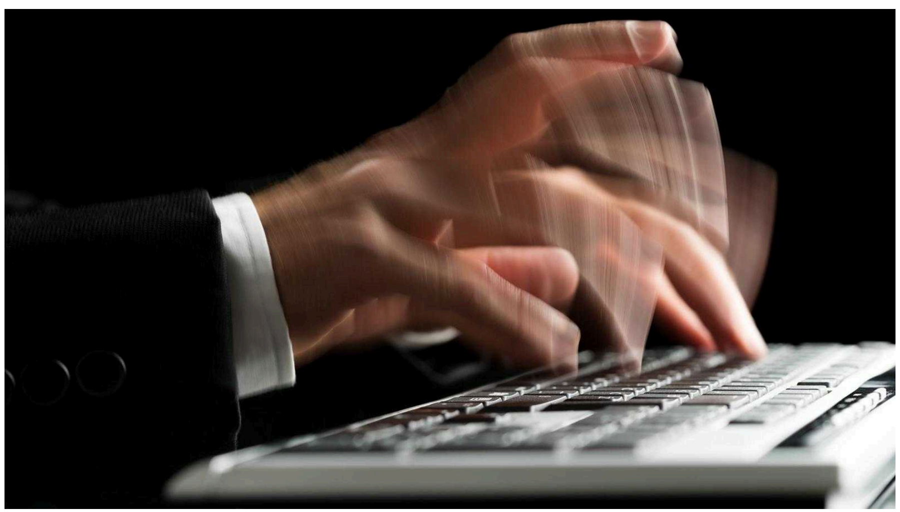
Homework and code handouts have the **R Markdown** format.



Write both text and R code in the **Environment**, then knit.

Code chunk shortcuts:

	Windows	**Mac**
Insert code chunk	Ctrl+Alt+I	Cmd+Opt+I
Run a single line of code	Ctrl+Enter	Cmd+Enter



Get used to these!!

Exercises | Solutions/Hints

- Add the line `months <- age * 12` to your Rmd.
- What happens when you run the chunk?
- What happens when you knit?

- Run the line `age <- 20` in your **console**.
- What happens when you knit?

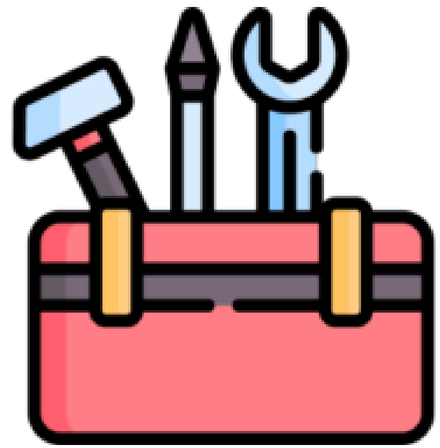
Thursday, 11:30pm

My chunks all run perfectly, but my file won’t knit! I’m getting a “object not found” error but it’s right there in the Environment.

Knitting a file creates a completely separate R instance - it can’t see your Environemnt tab. Make sure all objects are defined in the .Rmd itself, and don’t rely to much on the console!

We’ll be using the `tidyverse` package, an extra set of software for R. A package is like a toolbox.

Installation is like **buying** a toolbox



Loading is like **opening** a toolbox

Installing tidyverse

In your console, run

```r
install.packages("tidyverse")
```

You only need to do this once!

Additional troubleshooting

- `use a personal library`: pick yes
- `install from source`: pick no, then yes if it fails
- `compile`: pick no

Installation could take several minutes.

Looking ahead

In chapter 3, we'll learn the basics of writing code in R. If you want a head start, try to figure out on your own what the following lines of code do:

```r
favorite_numbers <- c(3, 1, 5, 6)
favorite_words <- c("apple", "mystical", "obsequious")

favorite_numbers[3]
favorite_numbers^2

str(favorite_words)
paste(favorite_words, "ly", sep = "")
```